Introduction to GitHub

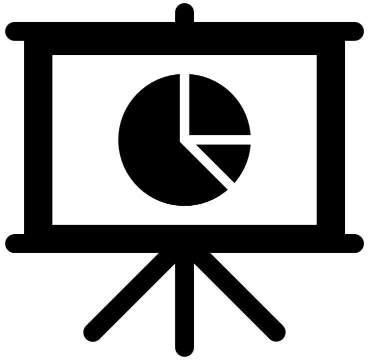
SISMID Lab

July 22 through July 24, 2026

A Human-AI Teaming Approach to Infectious Disease Modeling

Lecture Overview

General Housekeeping



Any slide with this icon contains the **overview** and **expected outputs** for the included tasks. All outputs listed on these slides should be uploaded to the course GitHub repository for review.

Any slide with this icon indicates that it is **time to complete a hands-on activity**. Carefully follow the instructions provided, complete the required task, and verify your results.

Task Overviews

Task	Description
Task 1: Forking GitHub	We will walk through how to **fork the course repository** on GitHub to create your own personal copy. You will create the fork, navigate to it, and open it so it's ready to work in during the lab.

Goal Output

From the activity (see previous slide), you should get your own **fork** of the lab's **GitHub repository** — a personal copy of the course repo hosted under your own GitHub account. By the end, your fork will be ready to open and work in during the lab.

Activity Output	Description
Your fork — github.com/your-username/course-repo	**A personal copy of the lab's GitHub repository** saved under your own GitHub account, which you fully control — browse files, make changes, and open pull requests throughout the lab.

A quick note before we dive in: This is an AI course — not an intro to GitHub. The tools we'll touch on are just that: tools, meant to support the real work, which is your modeling. We'll give you enough to feel confident and keep moving. And if you hit a snag along the way, don't worry — this is just a small piece of the picture, not the focus. If something sparks your curiosity or you want to go further, our door is always open.

GitHub Overview

What Is GitHub?

GitHub is a platform for storing, sharing, and collaborating on code and files. Think of it as a shared folder in the cloud — but with version history, quality checks, and structure.

Repository

A collection of folders on GitHub. Our class has one shared repository, made up of many folders.

Fork

Your own copy of the class repository on your GitHub account. You work here freely without affecting the original.

Commit

How you will upload your local changes from VS Code to GitHub. We will walk through this together after Activity 1 (*Data Cleaning*).

Pull Request

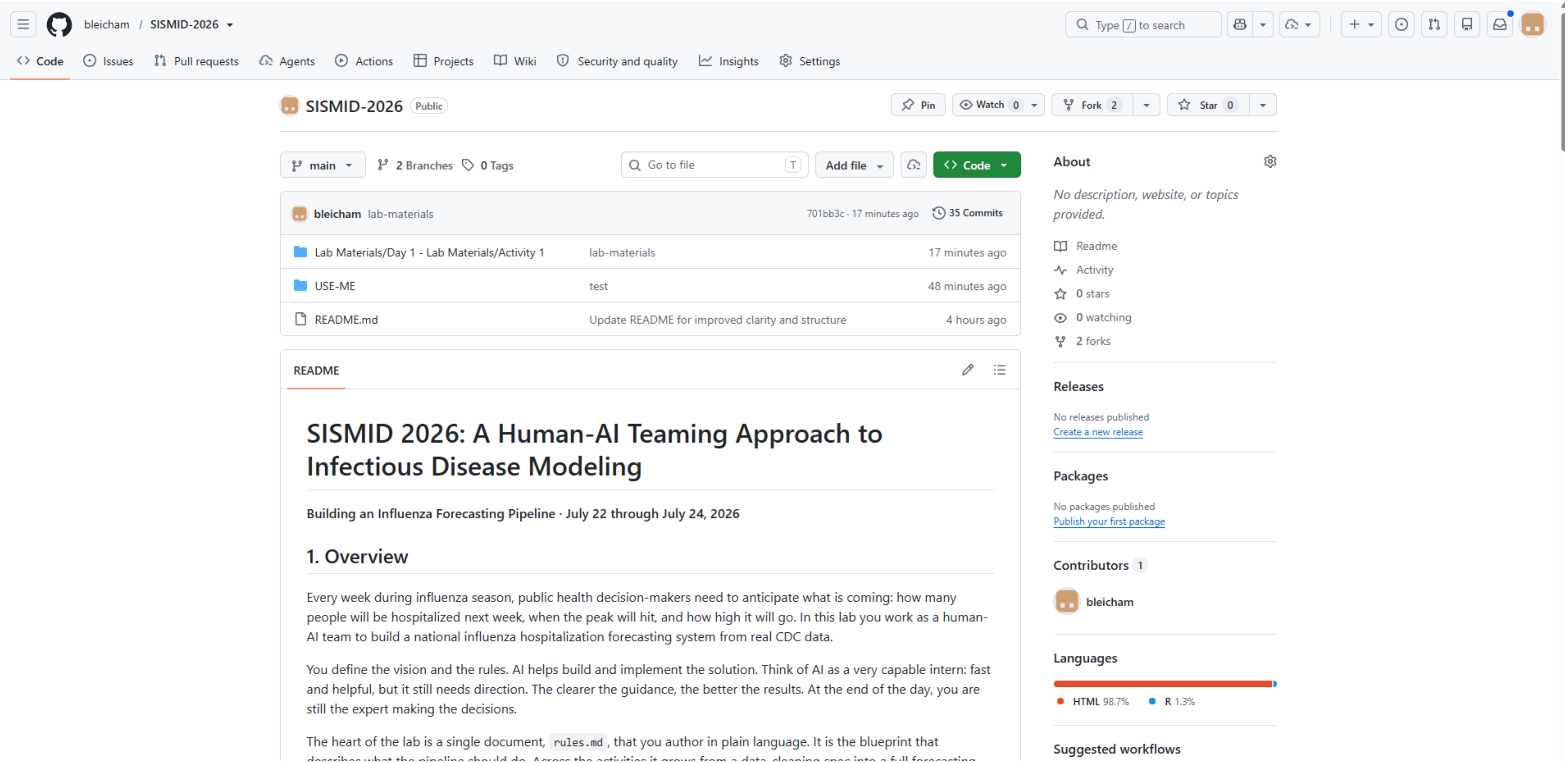
A request to merge your work into the main repository.

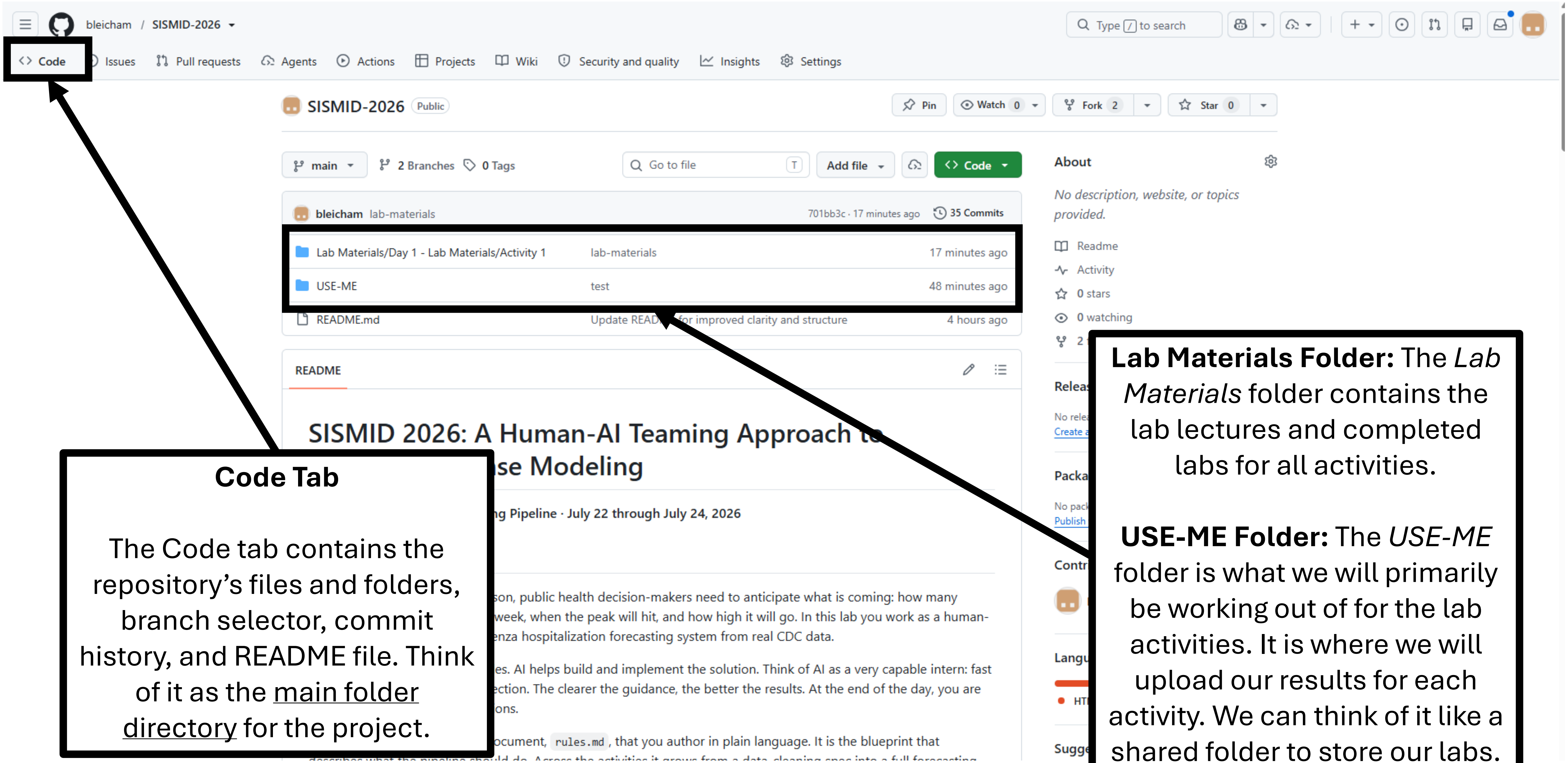
Key Tabs You'll Use

You don't need to know everything on GitHub for the labs — just these two tabs.

Code	Browse all files in the repository. This is the home page. You'll see the shared data folder and all student folders here. Click into folders to see their contents.
Pull Requests	This is where you turn in your work. After you commit your changes, open a pull request here. You can see where your work sits in the queue to be merged and follow its status along the way.

Our Course’s Repository



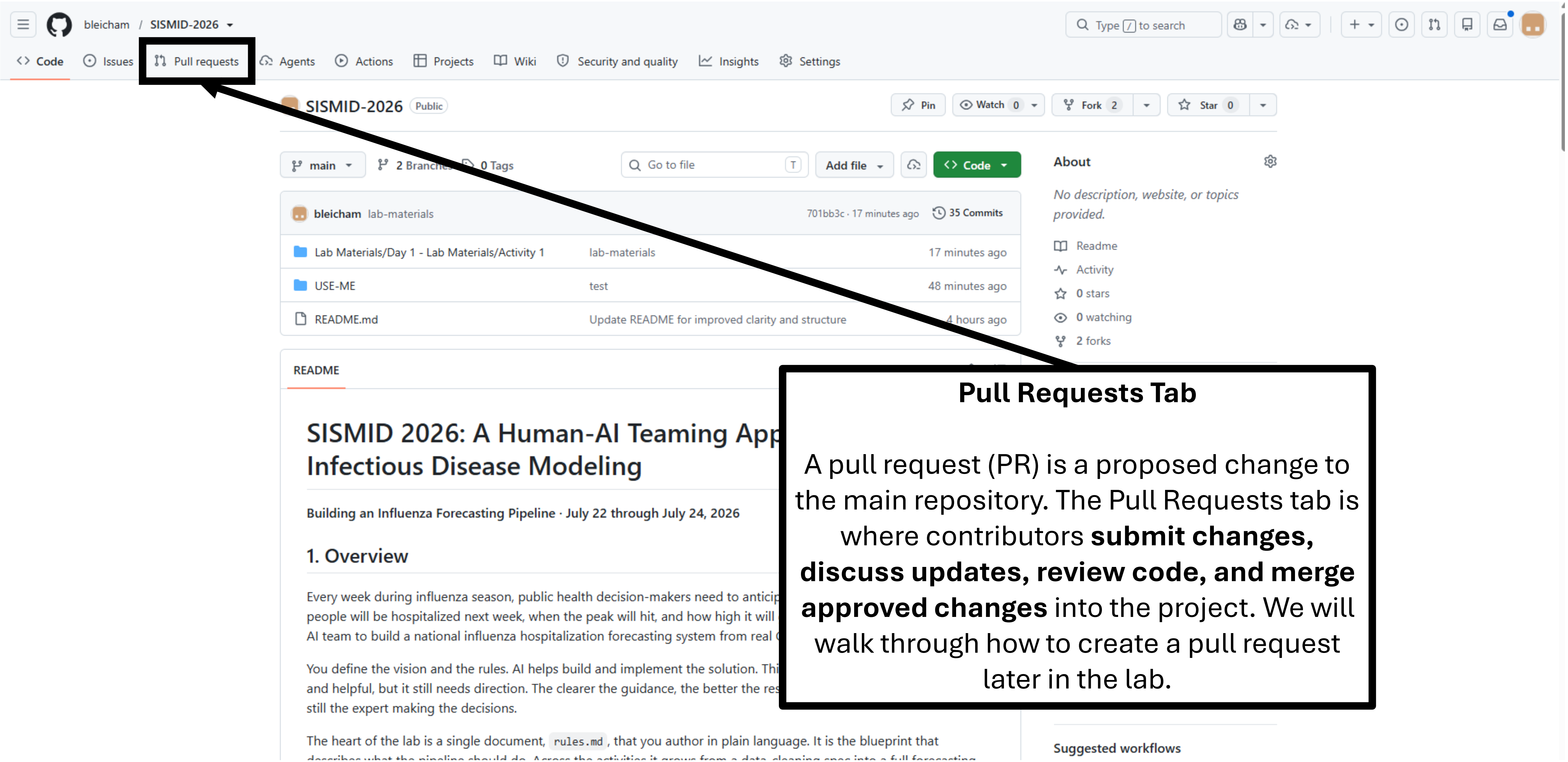


Code Tab

The Code tab contains the repository's files and folders, branch selector, commit history, and README file. Think of it as the <u>main folder directory</u> for the project.

Lab Materials Folder: The *Lab Materials* folder contains the lab lectures and completed labs for all activities.

USE-ME Folder: The *USE-ME* folder is what we will primarily be working out of for the lab activities. It is where we will upload our results for each activity. We can think of it like a shared folder to store our labs.



Pull Requests Tab

A pull request (PR) is a proposed change to the main repository. The Pull Requests tab is where contributors **submit changes, discuss updates, review code, and merge approved changes** into the project. We will walk through how to create a pull request later in the lab.

Lab Workflow: How will we use GitHub?

Work Locally (Fork)

Edit rules, generate code, run scripts in VS Code.

Commit to GitHub

Upload changes to your GitHub fork. Therefore, you will always have a GitHub-based repository with the code developed in this module.

Submit a Pull Request

Each time you make a change in your repository, you will also want to submit a 'pull request' to the course page.

You will repeat this cycle for each phase of the lab. Commit early and often — the agent gives you feedback each time and helps you keep a record of all your changes.

Activity: Forking GitHub

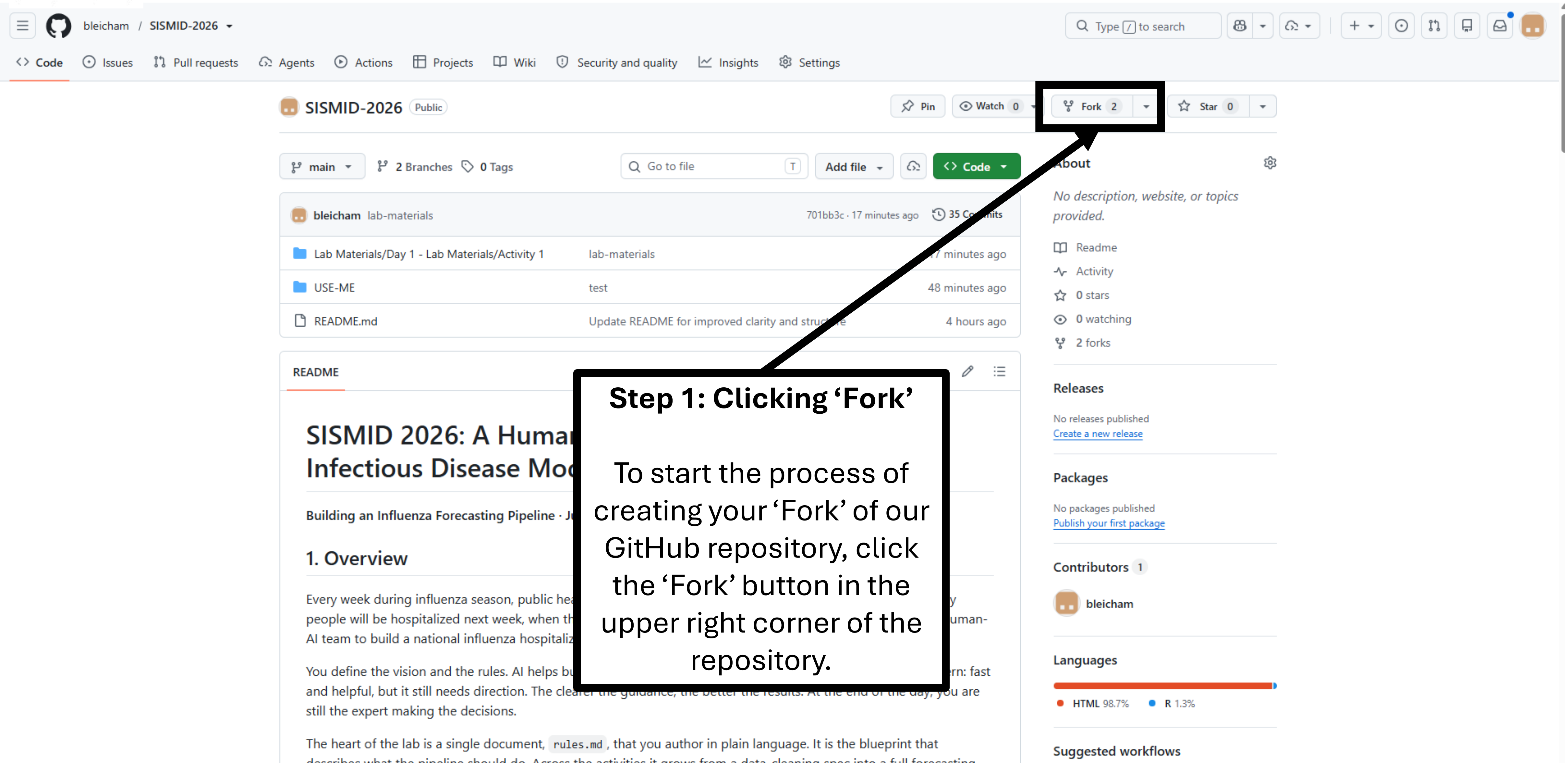
What Is a Fork?

A fork is **your own complete copy of a repository** on your GitHub account.

You can **freely edit your fork** without affecting the original.

When you're ready, you **submit a pull request** to merge your changes back.

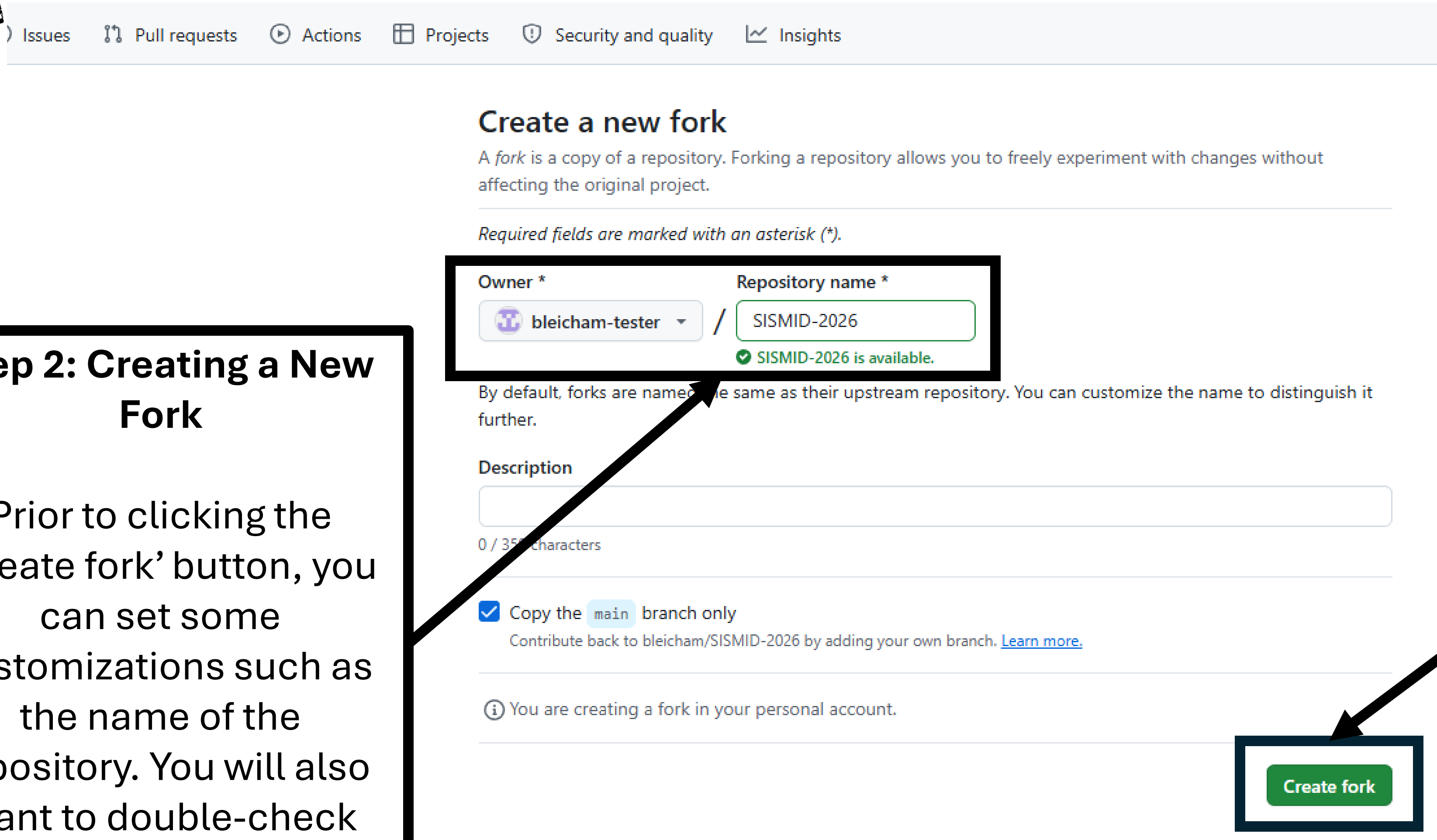
Starting with the Course Repository



YOUR TURN

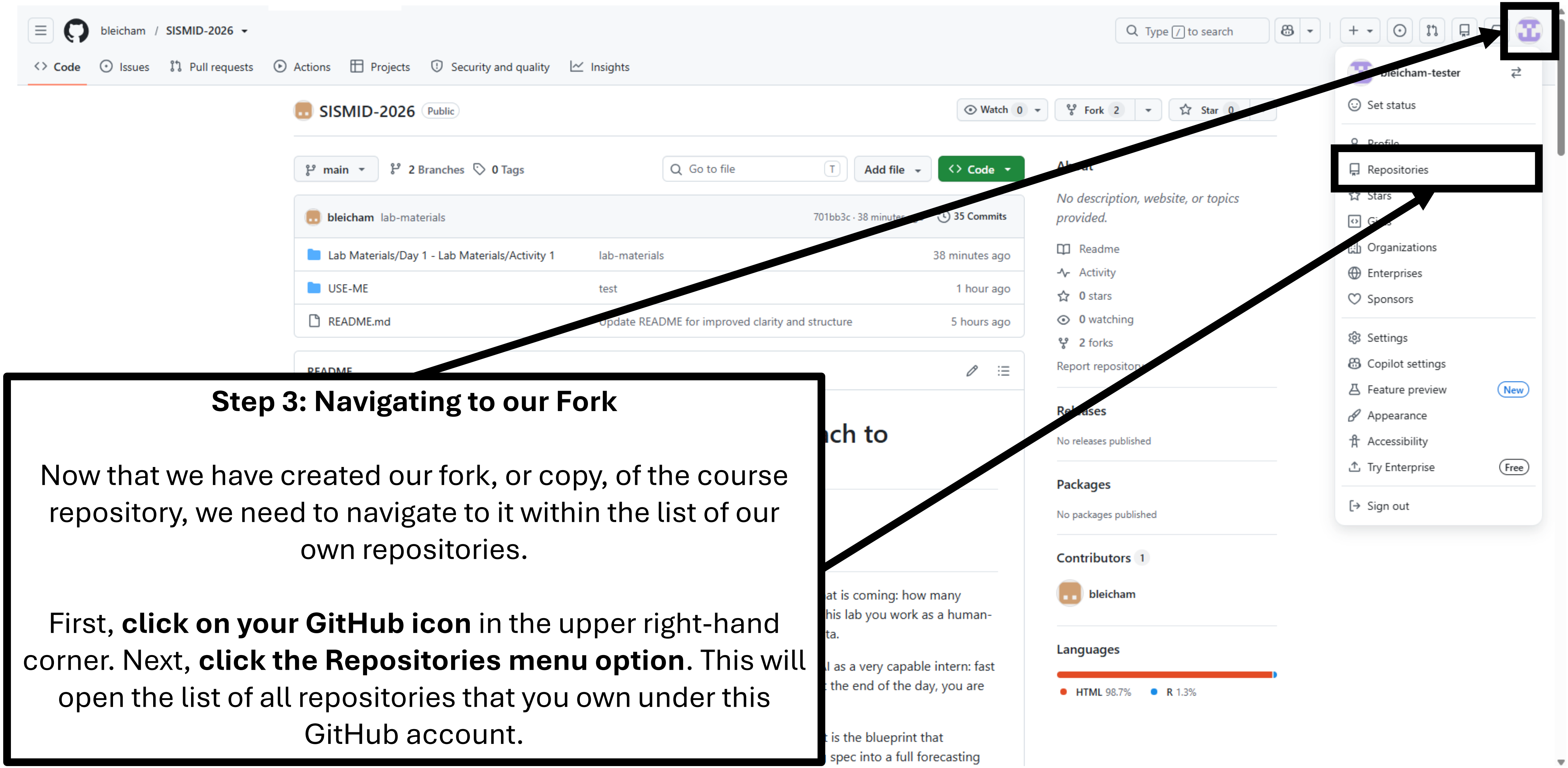
Step 2: Creating a New Fork

Prior to clicking the 'Create fork' button, you can set some customizations such as the name of the repository. You will also want to double-check that the correct `owner` is selected.



Once you are ready to copy the repository over to your account, click the 'Create fork' button.

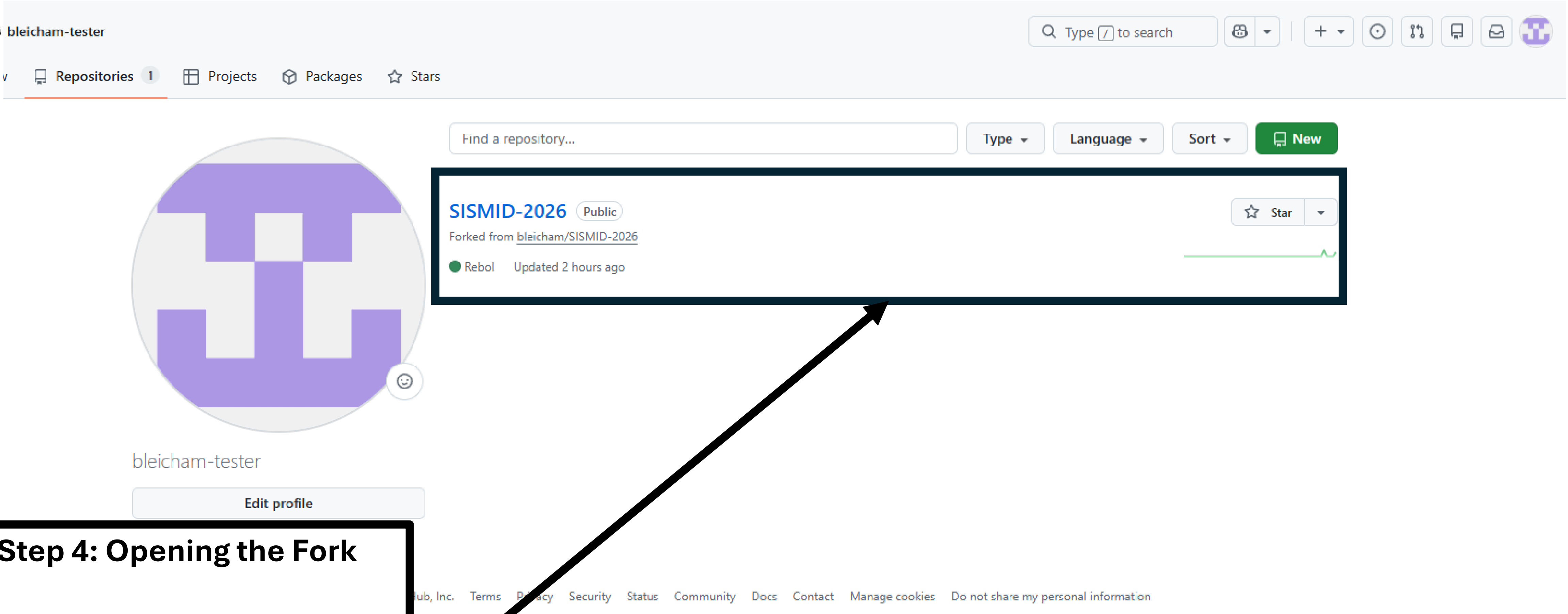
Navigating to our Fork



Step 3: Navigating to our Fork

Now that we have created our fork, or copy, of the course repository, we need to navigate to it within the list of our own repositories.

First, **click on your GitHub icon** in the upper right-hand corner. Next, **click the Repositories menu option**. This will open the list of all repositories that you own under this GitHub account.



Step 4: Opening the Fork

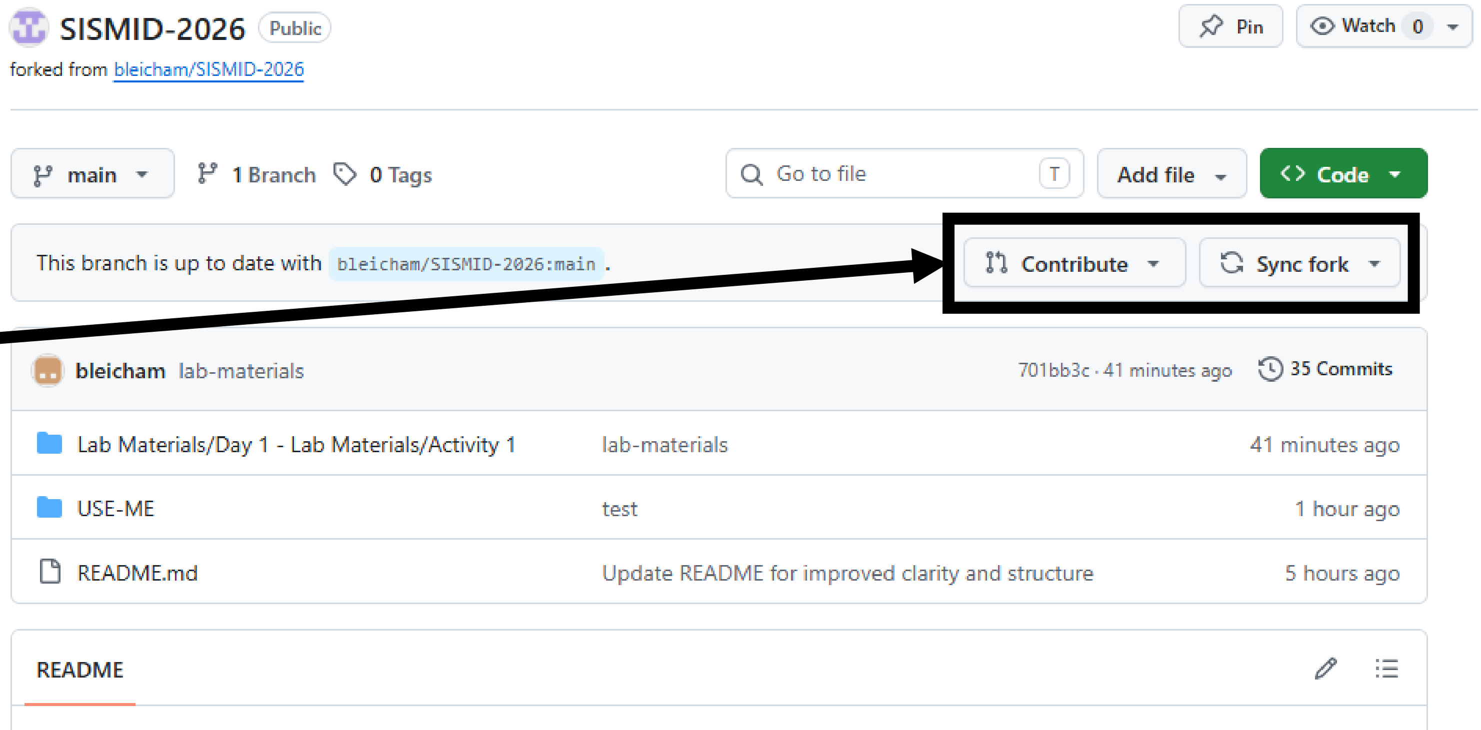
Now that we have navigated over to our own GitHub repositories page, we need to open the fork. To do so, click the title of the repository (*Blue title*).

YOUR TURN

You should now see your fork (personal copy) of the course GitHub repository in your own GitHub account.

At the top of the page, you should see a message indicating that your branch is up to date with the original repository, along with *Contribute* and *Sync fork* buttons. We will discuss different ways to contribute changes back to the original repository later in the lab.

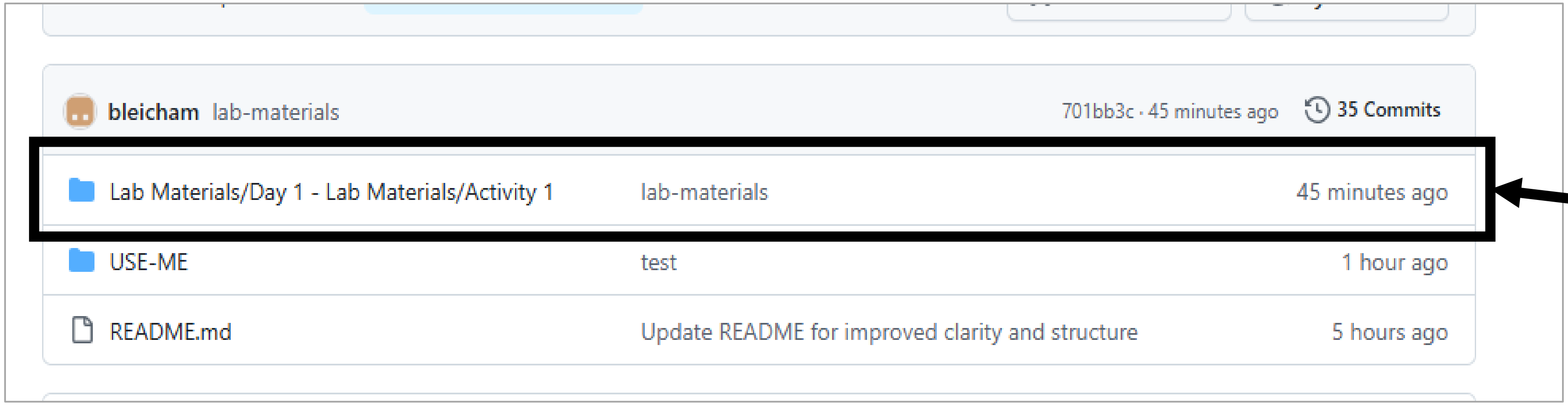
The *Sync fork* button allows you to pull in updates from the original repository, ensuring that your fork remains current with the latest changes.



YOUR TURN

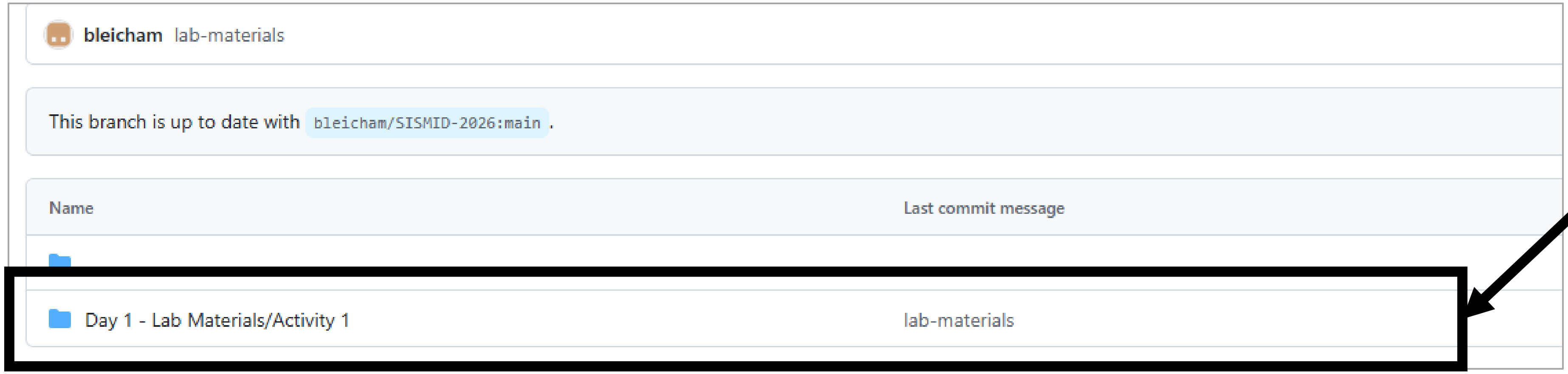
Exploring the *Lab Materials* Folder

Directory Structure



Step 1: Opening the *Lab Materials* folder

To open the *Lab Materials* folder, first **click** the *Lab Materials* folder row on the main GitHub page.



Step 2: Activity Files

When you first open the *Lab Materials* folder, you will only see the *Day 1* folder. Each day, a new folder will be added to this folder. To open, **click** the *Day 1* row.

YOUR TURN

Name	Last commit message
..	
DEMO	lab-materials
Hints and Answers	lab-materials
01_cleaning.pptx	lab-materials

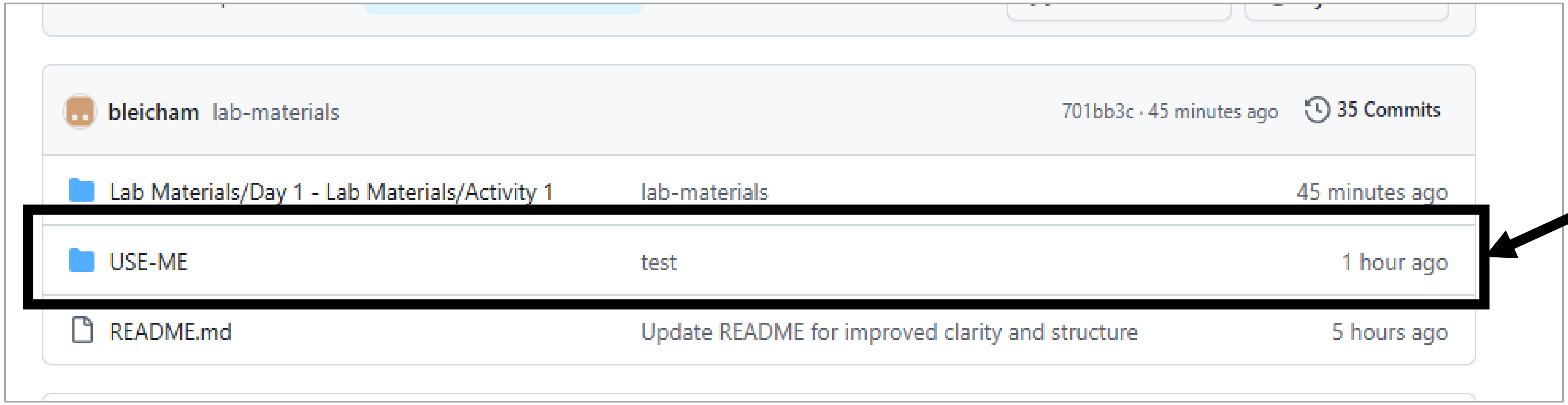
Step 3: Exploring the Activity folder structure

Although we're looking at the Day 1, Activity 1 folder, all Lab Materials folders follow the same format. Each contains a (1) Demo, (2) Hints and Answers, and (3) the PowerPoints for that activity.

(1) <u>Demo:</u> The starting point for the lab. If you fall behind or get stuck as we move between activities, you can grab the starting materials here instead of building off your own. **We will grab the contents of this folder in a bit (we will do this together).**
(2) <u>Hints and Answers:</u> The completed labs in VS Code along with the base R scripts.
(3) <u>PowerPoints:</u> What we will review for each lab. These are also available on the course website.

Exploring the *USE-ME* Folder

Directory Structure



Step 1: Opening the *USE-ME* folder

To open the *USE-ME* folder, first **click** the *USE-ME* folder row on the main GitHub page.

Step 2: Example Folder

As we move through the lab, you will see more folders show up here (one per person). We will cover the set-up in the next slide deck.

Final Directory Structure: Your Folder

```
NAMED FOLDER/
├── rules.md
├── AGENTS.md or agents.md
├── data/
│   └── Weekly Hospital Respiratory Data (HRD) Metrics by Jurisdiction.csv
└── output/
    ├── scripts/
    │   ├── 01_cleaning.R
    │   ├── 02_data_explore.R
    │   ├── 03_forecast.R
    │   ├── 04_evaluation.R
    │   └── 05_improvements.R
    ├── data/
    │   ├── 01_cleaning/
    │   ├── 02_data_explore/
    │   ├── 03_forecast/
    │   ├── 04_evaluation/
    │   └── 05_improvements/
    └── figures/
        ├── 01_cleaning/
        ├── 02_data_explore/
        ├── 03_forecast/
        ├── 04_evaluation/
        └── 05_improvements/
```